

Optimal configuration selection for Reconfigurable Manufacturing Systems

Ayman M. A. Youssef · Hoda A. ElMaraghy

Received: 11 September 2006 / Accepted: 11 May 2007 / Published online: 30 June 2007
© Springer Science+Business Media, LLC 2007

Abstract The selection of Reconfigurable Manufacturing Systems (RMS) configurations that include arrangement of machines, equipment selection, and assignment of operations, has a significant impact on their performance. This paper reviews the relevant literature and highlights the gaps that exist in this area of research. A novel “RMS Configuration Selection Approach” is introduced. It consists of two phases; the first deals with the selection of the near-optimal alternative configurations for each possible demand scenario over the considered configuration periods. It uses a constraint satisfaction procedure and powerful meta-heuristics, real-coded Genetic Algorithms (GAs) and Tabu Search (TS), for the continuous optimization of capital cost and system availability. The second phase utilizes integer-coded GAs and TS to determine the alternatives, from those produced in the first phase, that would optimize the degree of transition smoothness over the planning horizon. It uses a stochastic model of the level of reconfiguration smoothness (RS) across all the configuration periods in the planning horizon according to the anticipated demand scenarios. This model is based on a RS metric and a reconfiguration planning procedure that guide the development of execution plans for reconfiguration. The developed approach is demonstrated and validated using a case study. It was shown that it is possible to provide the manufacturing capacity and functionality needed when needed while minimizing the reconfiguration effort. The proposed approach can provide decision support for management in selecting RMS configurations at the beginning of each configuration period.

A. M. A. Youssef · H. A. ElMaraghy (✉)
Intelligent Manufacturing Systems Centre, Industrial & Manufacturing Systems Engineering,
Faculty of Engineering, University of Windsor, Room 204, Odette Building, 401 Sunset Avenue,
Windsor, ON, Canada N9B 3P4
e-mail: hae@uwindsor.ca

A. M. A. Youssef
e-mail: yousse5@uwindsor.ca

Keywords Optimization · Reconfigurable Manufacturing Systems · Reconfiguration planning · Reconfiguration smoothness · System configuration

1 Introduction

The history of manufacturing systems shows their evolution over the years in response to an increasingly dynamic and global market with a greater need for flexibility and responsiveness. Unpredictable market changes cause frequently varying manufacturing systems requirements. Reconfigurable Manufacturing Systems (RMS) were proposed to meet these requirements and provide a degree of capacity scalability and functional adaptability.

Most manufacturing industries now use a portfolio of Dedicated Manufacturing Lines (DML) and Flexible Manufacturing Systems (FMSs) to produce their products. RMS is intended to combine the high throughput of DML with the flexibility of FMS and react to changes quickly and efficiently (Koren et al. 1999). The characteristics of RMS and FMS are outlined and both paradigms are compared in (ElMaraghy 2005).

There are many aspects of manufacturing systems reconfiguration that present important research challenges. They include reconfiguration of the factory communication software, new machine controllers, building blocks and configuration of modular machines, modular processes and configuration of the production system (Mehrabi et al. 2000). The main focus of the research presented in this paper is the selection of system-level configurations.

A distinguishing feature of RMS is that its configuration evolves over time in order to provide the functionality and capacity needed, when it is needed. These configuration changes can be in the form of adding/removing machines to/from the system, adding/removing axes/spindles to/from machine tools, changing configuration of machine tools, changing the system layout, and changing the material handling systems.

It is desirable to change configuration when demand changes in order to minimize the unused capacity and functionality. In addition, there should be a high degree of reconfiguration smoothness between each two consecutive configurations in order to minimize the cost, time, and effort of reconfiguring the system. This paper proposes a novel “RMS Configuration Selection Approach” that selects RMS configurations according to the current situation, in terms of demand requirements, targeting the best achievable system performance levels while taking into consideration the smoothness of the anticipated reconfiguration process from one configuration to the next expected configuration. The proposed approach was implemented using MATLAB software and a toolbox was developed for this purpose.

2 Literature review

Wiendahl and Heger (2003) identified reconfigurability as one of five types of changeability of a manufacturing system. These types include, in addition to reconfigurability, changeoverability, flexibility, transformability, and agility. Makino

and Trai (1994) classified reconfigurable systems into statically and dynamically reconfigurable systems whereas ElMaraghy (2005) divided manufacturing systems reconfiguration activities into two types: hard and soft. Lee (1997, 1998) discussed reconfigurability in the design of products and manufacturing systems. Youssef and ElMaraghy (2006a) introduced a reconfiguration smoothness (RS) metric that reflects the ease and smoothness of transforming the system from one configuration to the next. Youssef (2006) further developed a stochastic model for evaluating the RS over a planning horizon and proposed a reconfiguration planning procedure that generates execution plans of reconfiguration. Kuo (2001) and Yamada et al. (2003) optimized the equipment layout assignment for RMS with the objective of minimizing the total transportation time. Abdi and Labib (2003, 2004) discussed strategic issues of selecting the system type and products grouping and selection using an analytical hierarchical process (AHP) model.

The following subsection provides an in depth review of the approaches that dealt with the selection of systems configuration in the RMS context in addition to other relevant research work.

2.1 Systems configuration selection approaches

Kimms (2000) presented a mathematical model formulation for the investment minimization of a flow line configuration, which was defined as the number of stages and the equipment in these stages that can handle multiple parts. It focuses only on the functional requirements of the system and does not consider the capacity requirements, which affect configuration selection decisions.

Cochran et al. (2001) integrated the axiomatic design approach with the concepts of several design frameworks and developed the Manufacturing System Design Decomposition (MSDD) approach. This approach can link the high-level objectives of a system to the operational level decision making process. However, it is difficult to apply this approach to the problem of multiple-aspect configuration selection for the following reasons: First, it does not consider optimization and enhancement of the manufacturing system performance because the main concern is to achieve a feasible system design within defined constraints and pre-specified objectives. In addition, it requires intensive and subjective human input to determine the Functional Requirements (FRs) and Design Parameters (DPs), which has to be repeated at the beginning of each configuration period in the context of RMS. Therefore, MSDD helps in system design synthesis with pre-specified objectives rather than seeking optimal performance, which makes it unsuitable for application to the problem at hand.

Matta et al. (2001) proposed a stochastic hierarchical approach for configuring automated production systems. The problem is decomposed into different sub-problems representing the production system. The level of detail and accuracy increases from the top of the hierarchy to the bottom, starting with the most crucial variables at the top levels, to be able to discard as many alternatives as possible. It is assumed that the impact of the different decision variables on the system performance is known before hand, which is difficult to predict. In addition, no mechanism for generating alternative configurations is offered.

Toenshoff et al. (2001) introduced a methodology, which uses evolutionary algorithms for investigating an optimized manufacturing system configuration. An evolution-based algorithm is used to create different alternative variations from a given starting configuration. Each of the new configurations is tested in a simulation environment and rated according to the user requirements. Only the best solutions for the required manufacturing system are used to continue with the evolution process until the ratings reach pre-defined stopping criteria.

Tesfamariam (2005) provides a comprehensive review of the methods that can be used in the different design stages for responsive manufacturing systems. The author identified quality, time, dependability, flexibility, and cost as important objectives for evaluating alternative system configurations and applied simulation based on system dynamics to investigate the suitability of an existing system configuration and its control policy to the manufacturing strategic objectives using a case study.

Xiaobo et al. (2000a) proposed a framework for a stochastic model of RMS. It involves three issues identified by the authors as the most important for which they proposed different solution approaches. These are: the optimal configurations in the design stage (Xiaobo et al. 2000b), the optimal selection policy in the utilization stage (Xiaobo et al. 2001a) and the performance measure in the improvement stage of these systems (Xiaobo et al. 2001b). Ohiro et al. (2003) proposed a modification that improved this work through involving the overall state of the system in choosing the best configuration. This research work does not clarify the information needed to define a configuration and assumes that a group of feasible configurations for each product family is available a priori.

Spicer et al. (2002) defined machining system configuration as the arrangement of the machines and the interconnections among them. They showed that, for the same number of machines, pure parallel configurations have the best throughput and scalability performance but with more quality streams than other types of configurations. Koren et al. (1998) used the same system configuration definition to demonstrate that the system configuration has a significant impact on six key evaluation criteria; investment cost of machines and tools, quality, throughput, capacity scalability, number of product types and system conversion time. Yang and Hu (2000) studied the effect of different configurations (parallel, series, etc.) on the system productivity using machine level reliability models for a six CNC machine manufacturing system. Maier-Sperdelozzi et al. (2003) studied the effect of different configurations on the manufacturing systems convertibility. Zhong et al. (2000) presented methodologies for evaluating system performance with respect to productivity, quality, scalability and convertibility for different machining system configurations. Maier-Sperdelozzi and Hu (2002) adapted the Analytic Hierarchy Process (AHP) for use in problems where manufacturing system configurations are selected considering multiple evaluation criteria.

Tang et al. (2004) introduced an approach for designing manufacturing lines that produce multiple parts. They utilized a Genetic Algorithm formulation to capture the configuration and task allocation for a multiple-parts line and used the minimal ratio of cost to throughput as the criterion for the fitness function. Tang et al. (2005) used this approach to prove that, for the same number of machines, the multiple

parts manufacturing system (MPMS) is less costly than the traditional single part manufacturing system (SPMS). This research work assumes that the designer predefines the number of production stages and that every part must visit all the stages in the system. These assumptions simplify the problem but affect the quality of the results by ignoring other system configuration possibilities.

Youssef and ElMaraghy (2006b) proposed a procedure for modeling and optimization of multiple-aspect, multi-part RMS configurations, the structure of which is that of flow lines allowing paralleling of identical machines in each production stage. The aspects considered in the RMS configurations include arrangement of machines, equipment selection and assignment of operations. They utilized Genetic Algorithms for the generation and assessment of the manufacturing systems configurations based on a mapping procedure that guarantees the feasibility of the generated alternatives. Youssef and ElMaraghy (2007) extended this work to incorporate the effect of machine availability using the Universal Generating Function (UGF) technique, based on the work done by Youssef et al. (2006), and added system availability as a second objective. They used Tabu Search (TS) as a second powerful optimization technique, with their model to compare the quality of the results with those produced using GAs and the run time when using both techniques. This work considers only a period of time with a fixed deterministic demand scenario and does not consider anticipations for either future demand scenarios or the corresponding effort of reconfiguring the system. These factors must be incorporated when selecting RMS configurations.

Son (2000) developed a methodology to design economical Reconfigurable Machining Systems (RmSs), given a deterministic demand scenario for the early stage of configuration design. This methodology generates configuration paths for changing demand by considering reconfigurations between demand periods, using a configuration similarity index, as well as the cost efficiencies for each demand period utilizing Genetic Algorithms (GAs). This work considers only single part demand scenarios, which is not sufficient when dealing with RMSs that are supposed to cope with flexibility in both functionality and capacity. In addition, the used index does not consider machine relocations, machine modules to be added/removed to/from the system and does not reflect the different levels of reconfiguration, which affects the influence of each component on the index evaluation. The Reconfiguration Smoothness (RS) metric proposed by Youssef and ElMaraghy (2006a) addresses these issues and reflects the activities associated with different levels of reconfiguration; market-level, system-level and machine-level.

Spicer (2002) developed a methodology to design scalable machining systems using an Integer Linear Programming (ILP) based partial enumerative procedure. It attempts to optimize the total life cycle cost of the system configuration including investment cost, operating cost in addition to reconfiguration cost. The author assumed that all used reconfigurable machine tools (RMTs) have identical machine bases and all the added or removed modules are identical. This assumption is unrealistic, since different types of RMTs can be equipped with different modules to be used for different process types such as milling, drilling, turning, and boring.

2.2 Summary of the literature review

A number of research issues and gaps exist regarding the configuration selection for RMSs. The following are some of these issues.

Most of the work done to date either handled the configuration problem from one configuration perspective, namely; physical machine arrangement (Koren et al. 1998; Zhong et al. 2000; Yang and Hu 2000; Maier-Sperdelozzi and Hu 2002; Spicer et al. 2002; Maier-Sperdelozzi et al. 2003) or dealt with the configuration as a parameter without defining it (Xiaobo et al. 2000a, b, 2001a, b; Ohiro et al. 2003). Both trends did not consider the automatic generation of feasible alternative configurations for different demand scenarios and considered a narrow pool of feasible configurations in the selection process. It is essential to consider other important aspects of configuration in addition to the automatic generation of feasible alternatives for the purpose of optimization. The complexity of the problem in this case requires powerful meta-heuristic tools such as GAs (Son 2000; Tang et al. 2004; Youssef and ElMaraghy 2006b, 2007) or TS (Youssef and ElMaraghy 2007).

A second issue is that most of the work done either focused on only the cost and economic benefits for performance evaluation (Son 2000; Xiaobo et al. 2000a, b, 2001a, b; Ohiro et al. 2003; Youssef and ElMaraghy 2006b) and did not consider other system evaluation criteria or coupled the cost with a second objective such as throughput (Spicer 2002; Tang et al. 2004) and system availability (Youssef and ElMaraghy 2007).

Another major shortcoming in most of the work done is neglecting the effect of the configuration selection on the smoothness of the subsequent reconfiguration process, which was only tackled by Son (2000) but with a basic configuration similarity model that needs major enhancements, Spicer (2002) but with an unrealistic reconfiguration cost model and Youssef and ElMaraghy (2006a); Youssef (2006) with a stochastic model of RS evaluation that is not yet incorporated in a configuration selection procedure.

An important drawback of the research work that considered the reconfiguration process as part of the configuration selection process (Son 2000; Spicer 2002) was dealing with the problem from a deterministic perspective. This is not sufficient especially when taking into consideration the anticipated demand and consequently the expected configuration and reconfiguration requirements.

Table 1 summarizes the above-mentioned issues for the most relevant research work in the area of configuration selection for RMS and highlights the gaps in this area of research. In conclusion, there is a need for a configuration selection approach that takes into consideration more than one perspective of the system configuration, involves the production of more than one part type simultaneously and involves stochastic analysis to anticipate the expected configuration requirements. This approach should be capable of generating a variety of feasible configuration alternatives, which can be accomplished by the use of a meta-heuristic optimization technique such as GAs or Tabu Search. These alternatives can be evaluated by the use of predetermined system evaluation criteria. In the meantime, the smoothness of the anticipated reconfiguration process between any two consecutive configurations

Table 1 Literature review summary for most relevant research work

	Aspects of configuration	Generation of configurations	System analysis measures	Optimization	Type of demand	Reconfiguration smoothness
Xiaobo et al. (2000a)	Not considered	Not considered	Cost	Classical	Stochastic	Not considered
Xiaobo et al. (2000b)	Not considered	Not considered	Cost	Classical	Stochastic	Not considered
Xiaobo et al. (2001a)	Not considered	Not considered	Cost	Classical	Stochastic	Not considered
Xiaobo et al. (2001b)	Not considered	Not considered	Cost	Classical	Stochastic	Not considered
Ohno et al. (2003)	Not considered	Not considered	Cost	Classical	Stochastic	Not considered
Yang and Hu (2000)	Physical arrangement	Not considered	Throughput	Not considered	Deterministic	Not considered
Koren et al. (1998)	Physical arrangement	Not considered	Multiple	Not considered	Deterministic	Not considered
Zhong et al. (2000)	Physical arrangement	Not considered	Multiple	Not considered	Deterministic	Not considered
Maier-Sperdelozzi and Hu (2002)	Physical arrangement	Not considered	Multiple	Not considered	Deterministic	Not considered
Spicer et al. (2002)	Physical arrangement	Not considered	Multiple	Not considered	Deterministic	Not considered
Maier-Sperdelozzi et al. (2003)	Physical arrangement	Not considered	Multiple	Not considered	Deterministic	Not considered
Tesfamariam (2005)	Layout	Not considered	Multiple	Not considered	Deterministic	Basic cost model
Youssef (2006)	Multiple	Not considered	Reconfiguration smoothness	Not considered	Stochastic	Considered
Youssef and ElMaraghy (2007)	Multiple	Considered	Cost and Availability	GAs and TS	Deterministic	Not considered
Tang et al. (2004)	Multiple	Considered	Cost and throughput	GAs	Deterministic	Not considered
Son (2000)	Multiple	Considered	Cost	GAs	Deterministic	Basic similarity metric
Spicer (2002)	Multiple	Considered	Cost and throughput	Classical	Deterministic	Unrealistic cost model

should be considered as a part of the configuration selection process. This has to be done stochastically to be able to handle the different future demand expectations. One more important aspect that needs to be incorporated in the selection of RMS configurations is the planning of the transition from one configuration to the next.

3 Basic assumptions and input/output description

3.1 Structure of the multiple-aspect RMS configuration

A reconfigurable manufacturing system should be capable of providing almost the exact capacity and functionality required to satisfy given demands for a group of products. Therefore, RMSs have characteristics similar to those of dedicated manufacturing systems within a configuration period (CP) because they are designed to be dedicated to the products for each CP with exact capacity and functionality. Both high production volume and high level of capacity scalability (the main characteristics of RMS) should be considered when deciding upon the RMS's basic structure.

Flow lines, being one form of RMS structures, can satisfy the high production volume requirements. In addition, flow lines can contain stages with multiple parallel stations (machines). This facilitates achieving the required scalability and synchronizes the different stages in order to maximize utilization of the available machines/stations. This would also reduce the effect of machines breakdown on the overall system performance. Therefore, the RMS configuration structure, used in this work, will be that of a flow line that allows paralleling of identical stations/machines with identical operation assignments in each production stage. This structure proved to be capable of satisfying the high production volume requirements of RMS and synchronizing the different production stages to maximize the overall system utilization (Youssef and ElMaraghy 2007). A similar structure was recommended in the final report of the “HIPARMS” (Highly Productive and Reconfigurable Manufacturing System) project sponsored by the Intelligent Manufacturing Systems (IMS) program (Fukaya 2004). Figure 1 shows an example of a selected configuration in a specific configuration period (CP) capable of producing two different part types within a family of parts.

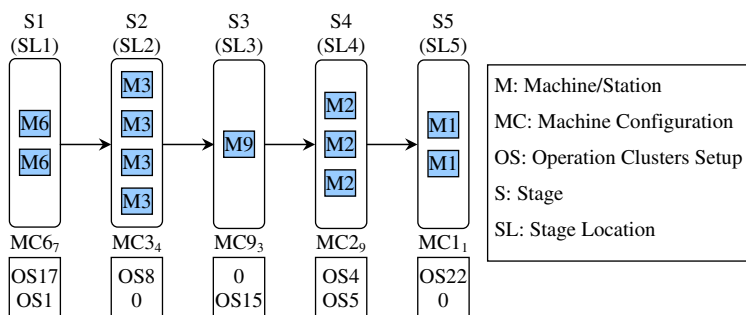


Fig. 1 Example of a selected RMS configuration in a specific configuration period (CP) (Youssef and ElMaraghy 2006a, b)

Therefore, a selected configuration is a series of stages each containing a group of parallel identical machines/stations. Each stage is represented by information such as stage location (relative to the available space for the flow line), machine/station type (stage type), and its selected machine configuration, number of machines/stations and the assigned operation clusters (operations). An operation clusters setup (OS) is a set of one or more operation clusters (OCs) that can be performed together on a specific machine with a specific configuration. An operation cluster (OC) is a set of operations (OPs), which are always machined together with a specific order due to different types of constraints such as logical or datum tolerance constraints. MC_{ij} stands for machine configuration j corresponding to machine/station i . Only one feasible machine configuration (MC) can be assigned to a machine/station (M) in a selected configuration. In Fig. 1, there are two rows of OSs each representing the OS assignments to different stages for one of the two part types to be produced and the zeros mean that the stage is not used for that specific part type.

3.2 Configuration periods (CPs)

The effort required to reconfigure the system according to anticipated demand scenarios has to be considered stochastically in selecting RMS configurations when dealing with such a dynamic and changing environment. A configuration period (CP) is a period with a fixed demand scenario (DS) that represents the product mix and volume requirements. More than one CP with more than one possible DS for each CP following the current CP are considered depending on available demand requirements information for each of these demand scenarios. This information includes the probability of occurrence of each DS in its corresponding CP in addition to the product mix (product types) and the production volume requirements for each product within this DS. The probabilities of occurrence of different DSs that belong to the same CP should add up to 1.

Figure 2 presents an example of demand scenarios (DSs) at each configuration period (CP) for the manufacturing of three parts A, B and C over four CPs. There is only one deterministic DS for the first CP while there might be more than one anticipated DS for the following CPs. In Fig. 2, DS_{ij} stands for demand scenario number j in configuration period number i , whereas P_{ij} stands for its probability of occurrence. The number in front of each product type represents the production volume requirement of that specific product type within its corresponding DS.

3.3 Input description

This subsection provides a brief description of the input parameters and information that are assumed to be available for the RMS Configuration Selection Approach.

3.3.1 Demand scenarios (DSs) information

These are the current demand scenario (DS_{11}) and the expected DSs for the following configuration periods (CPs) accompanied by their probabilities of

CP1			CP2			CP3			CP4		
DS ₁₁			DS ₂₁			DS ₃₁			DS ₄₁		
P ₁ = 1.00			P ₂₁ = 0.60			P ₃₁ = 0.45			P ₄₁ = 0.50		
Product	A	1000	Product	A	1000	Product	A	2000	Product	A	3000
	B	2000		B	3000		B	1500		B	1000
	C	0		C	0		C	1500		C	2000
			DS ₂₂			DS ₃₂			DS ₄₂		
			P ₂₂ = 0.40			P ₃₂ = 0.30			P ₄₂ = 0.50		
			Product	A	1500	Product	A	2000	Product	A	2000
				B	0		B	0		B	2500
				C	2500		C	2500		C	1500
						DS ₃₃					
						P ₃₃ = 0.25					
						Product	A	0			
							B	3000			
							C	1500			

Consider, in the following, one of the demand scenarios DS with a number of part types to be produced NP in a configuration period with duration T .

3.3.2 Parts processing information (OPs, OCs, OSs, and PGs)

OPs are the sets of operations required to produce each of the required parts. These must be accompanied by operations precedence graphs (PGs) that define sequential constraints between the different OPs and subsequently between different OCs and OSs. The following are data structures that capture information about OPs, OCs, OSs, and PGs:

$NOC[1,...,NP]$	vector representing the number of operation clusters (OCs) required to produce each part where $NOC(p)$ is the number of operation clusters (OCs) required to produce part p
$OPC_p[1,...,NOC(p)]$ [1,..., $NOC(p)$]	matrix to give information about the OPs of which each OC is composed for part $p \forall p = 1, \dots, NP$ where $OPC_p(i, y) = \begin{cases} 1, & \text{if } OP_y \text{ is a component of } OC_i \text{ for part } p \\ 0, & \text{otherwise} \end{cases}$
$NOS[1,...,NP]$	vector representing the number of operation clusters setups (OSs) for each part where $NOS(p)$ is the number of possible operation clusters setups (OSs) for part p
$NOS[1,...,NP]$	vector representing the number of operation clusters setups (OSs) for each part where $NOS(p)$ is the number of possible operation clusters setups (OSs) for part p
$OCS_p[1,...,NOS(p)]$ [1,..., $NOC(p)$]	matrix to give information about the OCs of which each OS is composed for part $p \forall p = 1, \dots, NP$ where $OCS_p(u, j) = \begin{cases} 1, & \text{if } OC_j \text{ is a component of } OS_u \text{ for part } p \\ 0, & \text{otherwise} \end{cases}$
$OSP_p[1,...,NOS(p)]$ [1,..., $NOS(p)$]	matrix to represent operation clusters setups precedence and feasibility of grouping relations of part $p \forall p = 1, \dots, NP$ where $OSP_p(u, v) = \begin{cases} 1, & \text{if } OS_u \text{ must be performed before } OS_v \text{ for part } p \\ 2, & \text{if } OS_u \text{ cannot be grouped with } OS_v \text{ (common OCs) for part } p \\ 0, & \text{otherwise} \end{cases}$ where u, v are the indices for operation clusters setups, $u, v = 1, \dots, NOS(p)$

3.3.3 Machines/stations (Ms) and machine configurations (MCs) information

This is the set of alternative reconfigurable machine/station types that are available/obtainable for use in the system. These Ms should be associated with the machine configurations (MCs) that can be used with each type and the corresponding cost and steady-state availability information. These MCs are accompanied by their corresponding feasible OCs and the number of removable modules (axes, spindles, etc.) that constitute each of them. In addition, each couple of MCs for the same machine/station (M) should be accompanied by the configuration distance between them in terms of the number of modules that have to be added/removed to/from any of them to obtain the other. The

following are the data structures that describe machines (Ms) and machine configurations (MCs) information:

NM	number of available/obtainable reconfigurable machine types
$NMC [1, \dots, NM]$	vector representing the number of possible machine configurations (MCs) that can be used with each machine type where $NMC(m)$ is the number of possible machine configurations (MCs) that can be used with machine type m and m is the index for machines, $m = 1, \dots, NM$
$CMC_m[1, \dots, NMC(m)]$	vector representing the initial cost of all possible MCs for machine $m \forall m = 1, \dots, NM$ where $CMC_m(c)$ is the Initial cost of machine configuration mc for machine m and c is the index for machine configurations, $c = 1, \dots, NMC(m)$. CMC_m includes cost of machine basic structure, modules for axes of motion, spindle modules and fixture modules
D	the depreciation rate for the equipment used in the configuration
I	annual interest rate
$AMC_m[1, \dots, NMC(m)]$	vector representing the machine steady-state availability of all possible MCs for machine $m \forall m = 1, \dots, NM$ where $AMC_m(c)$ is the steady-state availability of machine configuration c for machine m
$NRM_m[1, \dots, NMC(m)]$	vector representing the number of removable modules of all possible MCs for machine $m \forall m = 1, \dots, NM$ where $NRM_m(c)$ is the number of removable modules of machine configuration c for machine m
$DRM_m[1, \dots, NMC(m)]$ $[1, \dots, NMC(m)]$	matrix to represent the number of modules added and/or removed to/from machine m to change from one configuration to another $\forall m = 1, \dots, NM$ where $DRM_m(c, d)$ is the number of modules added to machine m to change from configuration c to configuration d or number of modules removed from machine m to change from configuration d to configuration c and c, d are the indices for machine configurations, $c, d = 1, \dots, NMC(m)$

3.3.4 Feasibility and operation time for each M-MC-OS combination

The feasibility and operation time information for a machine/station type (M) with machine configuration (MC) to perform an operation clusters setup (OS) should be provided. This enables the estimation of the production rate for this M-MC-OS combination in case it is feasible. The following are the data structure that represents production rate information:

$PROS_{p,m}[1, \dots, NOS(p)]$ $[1, \dots, NMC(m)]$	matrix to provide the production rate (in parts/h) of producing each possible operation clusters setup for part p using feasible configurations of machine $m \forall m = 1, \dots, NM \forall p = 1, \dots, NP$ where
	$PROS_{p,m}(u, c) = \begin{cases} \text{Production rate to produce OS}u \\ \text{using } Mm \text{ with } MCc \text{ for part } p, & \text{if feasible} \\ 0, & \text{if not feasible} \end{cases}$

3.3.5 Space limitations

The limitations regarding the space allocated for the flow line configuration include the length and width available for the configuration as specified by the system

designer. The length can be expressed by the number of stage locations (NSL), which determines the maximum number of stages. The width is expressed by the maximum number of parallel machines/stations within a stage (MMS).

3.3.6 Investment limitation

High level management normally sets the maximum initial investment in a manufacturing system configuration (MI) according to the budgetary constraints. This includes the cost of machines, axes, spindles and fixtures.

3.3.7 The configuration ($C0$) of $CP0$

This provides the full information that describes the configuration ($C0$), which was utilized in period $CP0$, the period prior to the period of interest ($CP1$). $C0$ is as exemplified in Fig. 1.

3.4 Output description

The output of the ‘‘RMS Configuration Selection Approach’’ is a group of sets of selected configurations. Each one of these sets consists of the selected configurations corresponding to all possible DSs over the system planning horizon including the configuration $C1$ selected for the current configuration period ($CP1$), the period of interest. Each of the selected configurations produced is exemplified in Fig. 1 and has the string representation illustrated in Fig. 3. The configurations, $C1$ s, corresponding to the current configuration period ($CP1$), the period of interest, are accompanied by the detailed execution plan of reconfiguration from the previous configuration ($C0$) to these configurations.

Figure 4 represents an IDEF0 model for the RMS Configuration Selection Approach that summarizes the inputs, outputs, mechanisms and control parameters. IDEF methods stand for ‘‘Integration DEFinition’’ methods. These are a group of standard modeling methods that can be used to describe operations in an enterprise. IDEF0, stands for IDEF for function modeling, which captures the first level of details. It is commonly used for representing the inputs, outputs, control parameters and mechanisms for any modeled activity or function. Refer to IEEE (1998) for further details.

4 RMS Configuration Selection Approach

This section provides a description of the proposed RMS Configuration Selection Approach, which is composed of two main phases.

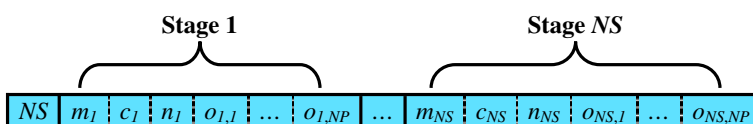


Fig. 3 String representation of a selected configuration (Youssef and ELMaraghy 2006b)

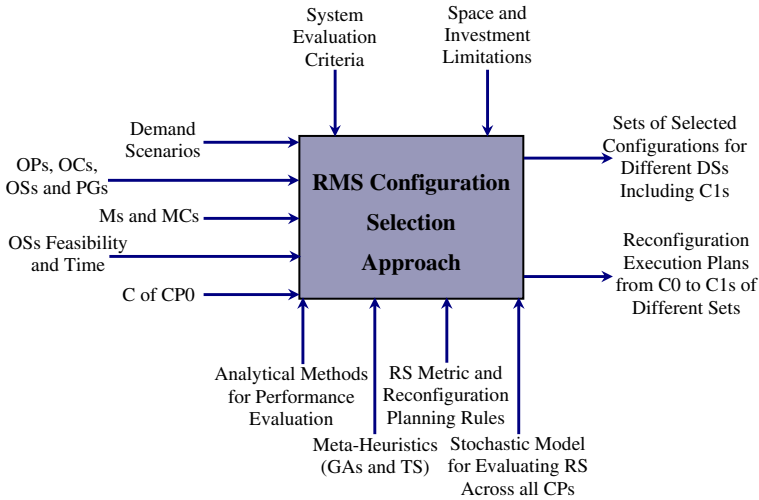


Fig. 4 IDEF0 model for the RMS Configuration Selection Approach

4.1 The first phase

This phase deals with the selection of the near-optimal alternative configurations for each individual demand scenario (DS) across all configuration periods (CPs) considered (see Fig. 2 as example of different DSs at each CP). This is an optimization process, which is governed by given system evaluation criteria regardless of the anticipated reconfiguration smoothness between consecutive configurations.

The optimization model proposed in (Youssef and ElMaraghy 2007) for including machine availability with no buffer capacity is adopted for this phase. It optimizes a utility/fitness function (Eq. 1) representing capital cost of configuration and, as a second criterion, system availability, which is assessed using the Universal Generating Function (UGF) technique that was first introduced by Ushakov (1986):

$$\text{Min } U = CC(NS, M, MC, NMS) + [1 - AV(NS, M, MC, NMS, OS)], \quad (1)$$

where AV is the system steady-state availability obtained using the UGF technique as described in (Youssef et al. 2006) and CC is the capital cost of the configuration defined as follows:

$$CC(NS, M, MC, NMS) = \sum_{s=1}^{NS} (n_s \times CMC_{m_s}(c_s) \times [1 - ((1-D)^T \times (P/F, I, T))]), \quad (2)$$

where

$$(P/F, I, T) = \frac{1}{(1 + I)^T}$$

is the present worth factor.

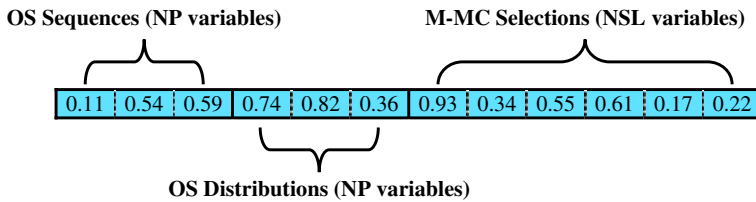


Fig. 5 String representation of the encoded configuration (Youssef and ElMaraghy 2006b)

Meta-heuristics, GAs and TS, are utilized for the generation and selection of the near-optimal group of configurations for each DS according to the chosen system evaluation criteria and the given constraints. A constraint satisfaction procedure is implemented through mapping from the discrete domain of variables (Fig. 3) to a continuous domain that guarantees the feasibility of all the generated configurations (Fig. 5). This coding scheme helps in the automatic generation of feasible configuration alternatives using the meta-heuristics which is followed by performance evaluation of these configurations.

The output of this phase is a predefined number of alternative configurations for each DS at each CP (see Fig. 1 as an example of a fully defined configuration). These selected configurations will be the near-optimal configurations according to the optimization performed by the meta-heuristic (GAs and TS). The number of these selected configuration alternatives (NC) will be the minimum of two values; a predefined number (default is 10) or the number of configurations the performance evaluation of which is within a predefined tolerance (default is 5%) compared to the best of these configurations. Using a predefined tolerance value (e.g. 5%) limits the number of alternative configurations that qualify to be considered in the second optimization phase to those that guarantee that further optimization does not sacrifice the first phase objectives. A predefined maximum number (e.g. 10) is used to reduce the computational complexity of the optimization process in the second phase by limiting its search space. Figure 6 illustrates an example of the output of the first phase for one of the considered DSs. In that figure, C_{ijk} represents alternative configuration number k for demand scenario DS_{ij} whereas NC_{ij} represents the number of alternative configurations for the same demand scenario.

The main reason for having more than one alternative for each DS is to provide the second phase of the procedure with a variety of good alternative configurations to choose from in order to achieve the best level of reconfiguration smoothness.

4.2 The second phase

This phase deals with determining which of the alternatives produced in the first phase for the demand scenario (DS) of the first configuration period (CP) (the period of interest) optimizes the degree of reconfiguration smoothness across all the configuration periods and the corresponding preliminary selected configurations for each of the possible DSs in the following CPs. This is accompanied by the execution plans to reconfigure the system from C0 to C1 for each set of configurations.

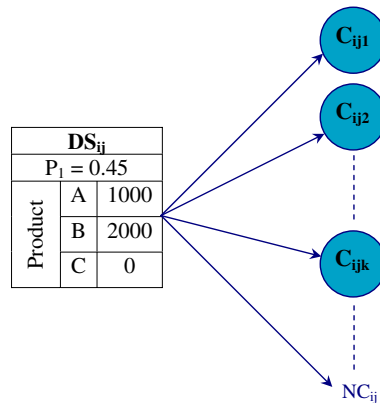


Fig. 6 Configuration alternatives for a demand scenario (DS_{ij})

4.2.1 Stochastic model of reconfiguration smoothness across (RSA)

The second phase of the RMS Configuration Selection Approach is based on a reconfiguration smoothness (RS) metric (Youssef and ElMaraghy 2006a) and a reconfiguration planning procedure (Youssef 2006). The RS metric gives a relative measure of the effort, time and cost of reconfiguring the system from one configuration to the next. The rule-based reconfiguration planning procedure helps in automatically determining the exact location of the different production stages in the system and thus developing execution plans for reconfiguring the system from one configuration to the next in a way that minimizes the reconfiguration effort. The reconfiguration planning rules, in the order of application to break possible ties, are as follows:

- Maximize the number of stage types that retain their original locations.
- Maximize the number of machines that retain their original locations.
- Minimize the number of empty stage locations between consecutive stages.
- Maximize the number of machines that keep their configurations.
- Maximize the number of machines that keep their operation clusters setup assignment.

A stochastic model (Youssef 2006) utilizes the RS evaluations to determine the reconfiguration smoothness across (RSA) corresponding to any candidate set of configurations corresponding to all the anticipated DSs at different CPs in the system's planning horizon. This model incorporates the RS evaluations of all pairs of possible consecutive configurations within this candidate set of configurations by multiplying the accumulated probability of occurrence of each possible pair of configurations by their respective RS evaluation. In addition, a predetermined value for relative importance is assigned to each CP.

Relative importance (RI) is a relative measure of the importance of each CP in the RSA evaluation. It reflects the closeness of each CP to the current point in

time and the degree of certainty/confidence off/in the anticipated stochastic information used in the RS evaluations for that CP. Accordingly the RI of each CP determines how its corresponding RS evaluations would influence the overall RSA evaluation and the configuration selection process in general. The value of RI for different CPs is left to the system designer to determine based on knowledge about the reliability of the anticipated information of the demand scenarios within these CP. The current CP (CP1), for example, should always be given the highest RI value because its corresponding demand scenario is deterministic and its selected configuration with the corresponding reconfiguration activities would be implemented immediately.

Accordingly, the RSA is obtained by summing up the pairwise RS evaluations of all possible consecutive pairs of configurations multiplied by their accumulated probabilities of occurrence multiplied by the RIs of the CPs to which these pairs belong. Figure 7 demonstrates an example of evaluating the RSA for a given set of selected configurations, where RI_i represents the relative importance assigned for CP_i in the RSA evaluation, C_0 is the configuration that was utilized in period CP_0 (the period prior to the period of interest, CP_1), P_{ij} is the probability of occurrence of DS_{ij} within CP_i , C_{ij} represents the configuration selected for demand scenario DS_{ij} and C_{ij} is in the form presented previously in Fig. 1. This example can be extended to evaluate sets of configurations for systems with different numbers of anticipated CPs and DSs in the planning horizon.

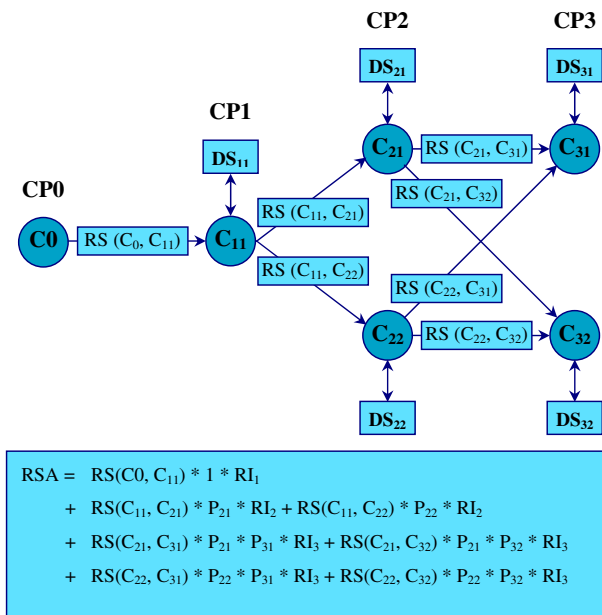


Fig. 7 An example of RSA evaluation (Youssef 2006)

4.2.2 Optimization of reconfiguration smoothness across (RSA) for a planning horizon

The optimization procedure starts with determining an upper bound for the RSA which is called reconfiguration smoothness limit (RSL). The RSL is the RSA corresponding to the set of configurations composed of the near-optimal configurations corresponding to the different DSs at different CPs (the near-optimal configuration for each DS among those generated from the first phase). This RSL is then used to constrain the optimization process, as it is not recommended to select a set of configurations with RSA inferior to that of the set of best configurations in addition to being inferior in terms of the criteria used in the first phase (cost and availability). A penalty function is used to ensure that the search tries to satisfy this constraint. If the RSA value exceeds the maximum allowable value RSL, a penalty value of 100 multiplied by the exceeded value is added to the objective function value, which represents the original RSA.

The next step is to perform optimization in order to generate and select a predefined number of near-optimal sets of configurations according to the RSA evaluation without violating the RSL. In this optimization process, the different demand scenarios (DSs) are treated as variables for which the domains of values are the alternative configurations ordered by their fitness value obtained from the first phase for each DS. Figure 8 illustrates a string representation of a selected set of configurations, where each digit represents the order of the configuration selected from those produced in the first phase corresponding to each possible DS at each CP.

This is a discrete optimization problem for which the search space has an order that is exponential in the total number of demand scenarios over all the considered configuration periods. For example, if the predefined number of alternative configurations for each DS at each CP was equal to 10, then the size of the search space is 10 to the power of $\sum_{cp=1}^{NCP} NDS(cp)$, which would explode exponentially in case of considering a large number of anticipated demand scenarios. Accordingly, the use of a powerful meta-heuristic optimization technique is crucial to overcome the computational complexity of the problem.

4.2.3 Optimization using Genetic Algorithms (GAs) and Tabu Search (TS)

Genetic Algorithms (GAs) technique (Holland 1975), which is broadly used in solving multi-modal and exponential problems, was chosen for solving the above-described problem of discrete optimization of the RSA.

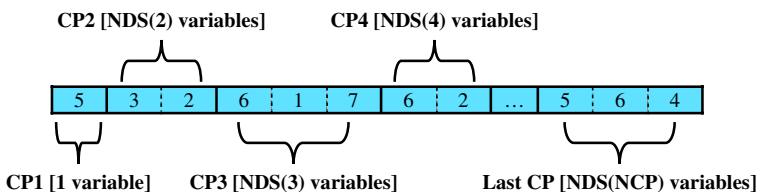


Fig. 8 String representation of a selected set of configurations

A second powerful meta-heuristic global optimization technique was needed to compare the quality and computation time of the results obtained by GAs. Tabu Search (TS) (Glover 1986), a well-established technique that proved to be successful in many applications (Glover and Laguna 1997), was chosen. A variant of TS that proved to be very efficient in solving combinatorial problems, namely; Reactive Tabu Search (RTS) (Battiti and Techiolli 1994), was utilized in the optimization of the RSA. Appendix A provides a brief idea about the use of basic Tabu Search (TS) (Glover and Laguna 1997) and its variant, RTS.

4.2.4 Output of the second phase

The output of the second phase, and accordingly the overall RMS Configuration Selection Approach, is a number of candidate configurations for the current (first) CP (the period of interest). Each of these candidate configurations will be accompanied by a preliminary selection of a combination of configurations across all CPs that optimizes the RSA evaluation. The number of these selected sets of configurations (NSC) will be the minimum of two values; a predefined number (default is 10), to limit the number of reported sets, or the number of configuration sets the performance evaluation of which is within a predefined tolerance (default is 5%) compared to the best set, to ensure the quality of these sets in terms of their RSA value. In addition, and for each of these sets, the execution plans for reconfiguring the system from C0 to C1 of that set is developed according to the reconfiguration planning rules. Figure 9 presents an example of a selected set of configurations for a system with only two CPs in its planning horizon.

The generated sets of configurations not only have near-optimal RSA evaluations but are guaranteed to have evaluations for the criteria used in the first phase (cost and availability) that are within a very small tolerance compared to the near-optimal configurations. Figure 10 summarizes the overall approach.

5 Case study

In order to demonstrate the application of the developed RMS Configuration Selection Approach, a case study is adapted from (Youssef and ElMaraghy 2006a, c) based on an example part (CAM-I,¹ 1986 test part ANC-101) that is widely used in the literature. Figure 11 shows parts ANC-101 and ANC-90 that belong to the same products family. Part ANC-90 is a part instance that has five fewer features than those present in part ANC-101 as well as differences in their common features such as the number of holes in the top surface of both parts. Table 2 provides the description and initial cost of all the resources (M-MC combinations) that can be used in the system. The depreciation rate for these machines is assumed to be 10%. All the data for the two parts and their machining processing information using the resources provided in Table 2 can be found in Appendix B. The weights used in evaluating the different components of the RS metric are the same as those used in

¹ CAM-I: Computer Aided Manufacturing—International.

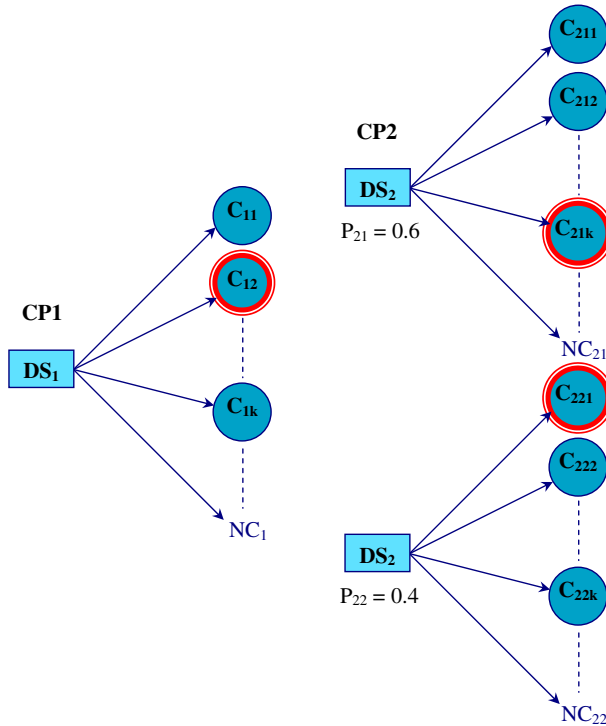


Fig. 9 A selected set of configurations for a system with only two CPs

(Youssef and ElMaraghy 2006a). Implementation of the case study was carried out using the developed MATLAB toolbox, RMS-Configurator, on a Pentium 4 3.4 GHz PC with 1.0 GB memory.

Consider the case of having an existing configuration C0 (Fig. 12) that was utilized in the previous configuration period (CP0), the period prior to the period of interest (CP1), and having five CPs in the planning horizon of a RMS with different numbers of anticipated DSs. NMS, used in the configuration representation illustrated in Fig. 12, represents the number of identical parallel machines in each production stage. Table 3 provides all the information regarding the five CPs and their anticipated DSs including the duration of each CP, the relative importance (RI) of each CP regarding the RSA evaluation, the product mix and production volume requirements for each DS in each CP. The annual interest rate is assumed to be 12%. The system designer specified the maximum number of stages to be 10 and the maximum number of parallel machines per stage to be 8. The maximum allowable budget for initial investment is 60 million US Dollars.

5.1 Results of the first phase

The first phase of the RMS Configuration Selection Approach had to run first in order to provide the second phase with alternative configurations for each demand

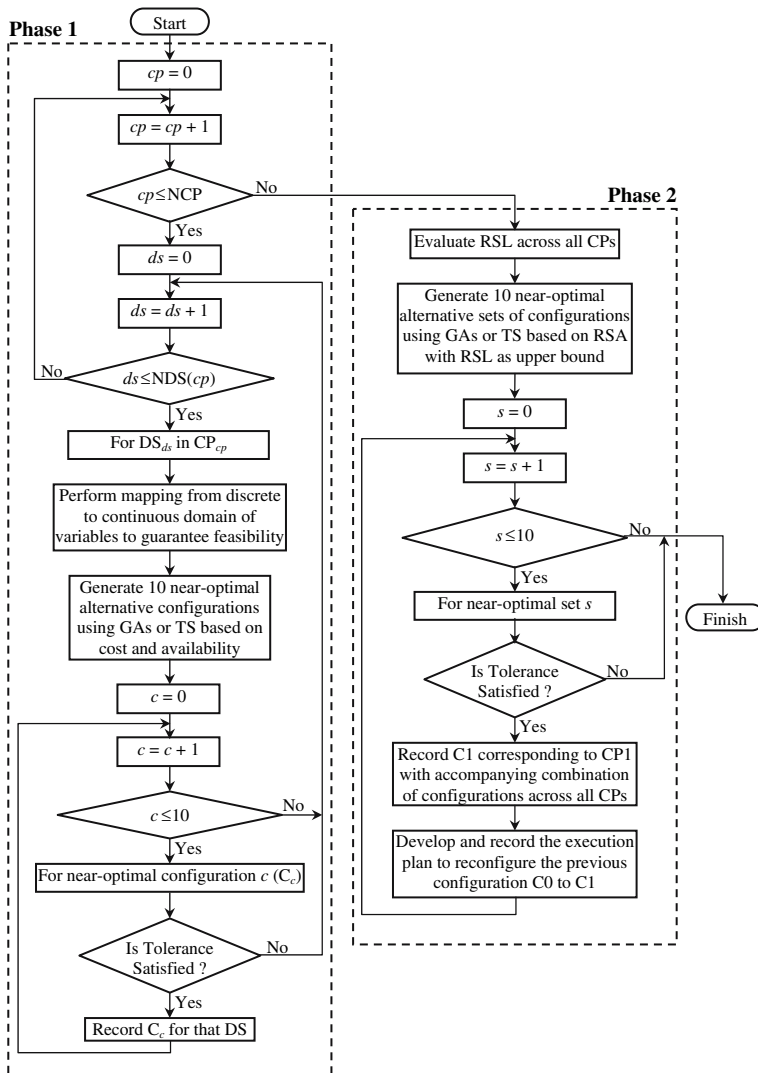


Fig. 10 The overall RMS Configuration Selection Approach

scenario (DS) in each configuration period (CP) within the planning horizon. The validated continuous optimization model and GAs technique with same parameters (Table 4) were adopted from (Youssef and ElMaraghy 2007) and utilized for generating the near-optimal configurations according to cost and availability for each DS in each CP. Real-Coded GAs (Herrera et al. 1998) were preferred to TS as they obtained better results in (Youssef and ElMaraghy 2007). The number of configurations generated for each DS was the minimum of 10 or the number of configurations within 5% tolerance limit compared to the best of these configurations.

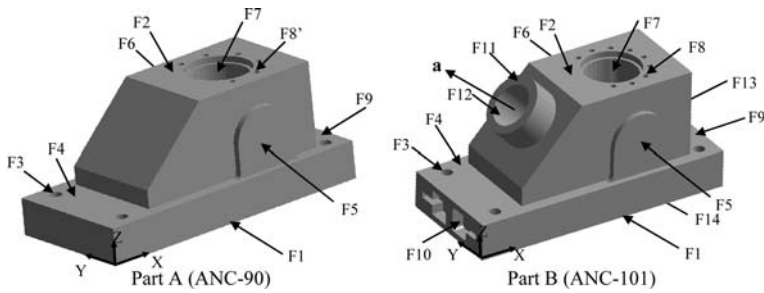


Fig. 11 Parts used in the case studies

Table 2 Available/obtainable resources description and cost

Machine (M)		Machine configuration (MC)		Initial cost (in 1000 of USD)	Steady-state availability	Number of removable modules
Code	Description	Code	Description			
M1	Reconfigurable horizontal machining centre	MC1 ₁	3-axis with 1 spindle	860	0.92	3
		MC1 ₂	3-axis with 2 spindles	1140	0.90	4
		MC1 ₃	3-axis with 3 spindles	1420	0.88	5
		MC1 ₄	3-axis with 4 spindles	1700	0.86	6
		MC1 ₅	4-axis with 1 spindle	1010	0.90	4
M2	Reconfigurable drilling press	MC2 ₁	1 spindle	385	0.94	1
		MC2 ₂	2 spindles	555	0.92	2
		MC2 ₃	3 spindles	725	0.90	3
		MC2 ₄	4 spindles	895	0.88	4
M3	Reconfigurable boring machine	MC3 ₁	1 spindle	285	0.92	1

S (SL)	1 (3)	2 (4)	3 (5)	4 (6)	5 (7)	6 (8)
M	1	1	1	2	2	1
MC	5	3	5	3	1	2
NMS	4	6	5	2	6	2
OS _A	1	14	5	6'	3	0
OS _B	9	5	13	6	3	11

Fig. 12 Configuration C0 of CP0

Table 3 Case study information

Configuration periods information			Demand scenarios information		
CP#	Duration (in years)	RI	DS#	Demand requirements (in parts/h)	
				Part A	Part B
CP1	1.5	0.45	DS ₁₁	120	180
CP2	1.0	0.25	DS ₂₁	220	0
			DS ₂₂	180	120
CP3	1.2	0.15	DS ₃₁	120	180
			DS ₃₂	180	180
			DS ₃₃	0	200
CP4	1.5	0.10	DS ₄₁	0	220
			DS ₄₂	120	120
CP5	1.3	0.05	DS ₅₁	150	150
			DS ₅₂	150	120
			DS ₅₃	120	150
			DS ₅₄	250	0

Table 4 Parameters used in real-coded GAs (Youssef and ElMaraghy 2007)

Parameter	Value
Population size	200
Number of generations	700
Number of times of cross-over, application (arithmetic cross-over, simple cross-over, and heuristic cross-over)	Six times for each operator
Number of times of mutation application (uniform mutation, boundary mutation, non-uniform mutation, and whole non-uniform mutation)	Twelve times for each operator

Figure 13 and Table 5 provide a sample of the results obtained at the end of the first phase. Figure 13 illustrates only the near-optimal configuration obtained corresponding to each DS in each CP and Table 5 provides the performance characteristics of these configurations, where C_{ij} is the near-optimal configuration selected for DS_{*j*}. The remaining configurations that were generated at the end of this phase to be considered for the second phase optimization process were not possible to report due to space restrictions.

In Table 5, the expected production rate of each part represents the expected value of the actual production capacity (possible production rate) of the manufacturing system for that part obtained using the UGF after considering machine availability (up/down times). System utilization (Eq. 4) is the summation, for all parts produced by the system, of the ratio of the system demand requirements (system output) (D) to the expected production rate (EPR) for each part (Youssef and ElMaraghy 2007):

Configuration C₁₁							
S	1	2	3	4	5	6	7
M	1	1	1	1	1	2	1
MC	2	3	4	5	2	5	2
NMS	3	3	2	7	2	4	6
OS_A	1	0	14	5	0	3	6'
OS_B	1	15	13	5	11	9	6

Configuration C₂₁				
S	1	2	3	4
M	1	1	1	2
MC	3	3	5	4
NMS	2	2	5	2
OS_A	1	15	5	6'

Configuration C₂₂							
S	1	2	3	4	5	6	7
M	1	1	1	1	2	1	1
MC	3	3	5	5	3	3	3
NMS	2	3	4	7	4	2	2
OS_A	1	15	0	5	6'	0	0
OS_B	1	14	9	5	6	13	12

Configuration C₃₁							
S	1	2	3	4	5	6	7
M	1	1	1	2	1	2	1
MC	3	3	5	1	5	4	2
NMS	2	2	7	4	4	4	2
OS_A	1	2	5	0	7	6'	4
OS_B	1	14	5	3	9	6	11

Configuration C₃₂							
S	1	2	3	4	5	6	7
M	1	1	1	1	1	1	2
MC	3	4	5	3	5	3	3
NMS	2	2	7	3	8	2	6
OS_A	0	0	1	15	5	0	6'
OS_B	1	14	5	12	9	13	6

Configuration C₃₃							
S	1	2	3	4	5	6	7
M	1	1	1	1	2	1	1
MC	2	5	5	5	2	1	1
NMS	2	7	3	5	5	2	1
OS_B	1	17	9	5	6	11	11

Configuration C₄₁						
S	1	2	3	4	5	6
M	1	1	1	1	2	1
MC	2	5	5	5	2	3
NMS	2	5	4	5	6	2
OS_B	1	16	9	5	6	12

Configuration C₄₂						
S	1	2	3	4	5	6
M	1	1	1	2	1	1
MC	2	5	5	2	5	3
NMS	2	6	6	5	3	2
OS_A	1	15	5	6'	0	0
OS_B	1	16	5	6	9	12

Configuration C₅₁							
S	1	2	3	4	5	6	7
M	1	1	1	2	1	2	1
MC	3	5	5	3	5	3	2
NMS	2	7	4	2	7	4	2
OS_A	1	15	0	0	5	6'	0
OS_B	1	16	9	3	5	6	11

Configuration C₅₂							
S	1	2	3	4	5	6	
M	1	1	1	1	1	2	
MC	2	5	5	3	5	3	
NMS	3	6	6	2	4	4	
OS_A	1	15	5	0	0	6'	
OS_B	1	16	5	12	9	6	

Configuration C₅₃							
S	1	2	3	4	5	6	7
M	1	1	2	1	1	1	2
MC	2	5	3	5	5	2	2
NMS	2	6	2	7	5	2	6
OS_A	0	1	0	15	5	0	6'
OS_B	1	16	3	5	9	11	6

Configuration C₆				
S	1	2	3	4
M	1	1	1	2
MC	2	3	5	3
NMS	2	2	6	3
OS_A	1	15	5	6'

Fig. 13 Near-optimal configurations obtained in the first phase for each DS in each CP

$$\text{System utilization} = \sum_{p=1}^{NP} \frac{D(p)}{EPR(p)}. \quad (4)$$

The computation time used in the first phase by the developed MATLAB toolbox to generate the near-optimal group of alternative configurations for each DS in each CP was 12 h with an average of 1 h per DS. The alternative configurations generated in the first phase by the optimization model for each DS in each CP within the

Table 5 Performance characteristics for best configurations obtained in first phase

Configuration	Capital cost (in millions of US dollars)	Initial investment (in millions of US dollars)	System availability (in %)	Expected production rate for Part A (in parts/h)	Expected production rate for Part B (in parts/h)	System utilization (in %)
C ₁₁	7.7746	27.8	74.161	341	280	99.3
C ₂₁	2.4593	12.52	54.291	221	N/A	99.4
C ₂₂	5.2623	26.79	59.022	341	256	99.5
C ₃₁	6.239	27.03	74.884	340	279	99.9
C ₃₂	7.58	32.84	77.024	400	332	99.2
C ₃₃	5.0607	21.925	51.696	N/A	200	99.9
C ₄₁	6.3175	22.59	60.779	N/A	220	99.8
C ₄₂	6.4448	23.045	57.121	282	210	99.6
C ₅₁	6.8422	27.65	72.137	336	276	98.9
C ₅₂	6.2656	25.32	53.037	296	246	99.5
C ₅₃	6.81	27.52	58.066	269	277	98.9
C ₅₄	3.3048	13.355	40.502	257	N/A	97.4

planning horizon are now provided to the second phase in order to select from them the sets of configurations that minimize the required effort of reconfiguration along the planning horizon.

5.2 Results of the second phase

Both GAs and RTS, described in Sect. 4.2.3, were applied in selecting the sets of configurations, from those produced in the first phase, that optimize the reconfiguration smoothness across (RSA) value along the designated planning horizon, as described in Sect. 4.2.2. Both optimization techniques are capable of generating a number of near-optimal sets of configurations in each run. This number is the minimum of 10 or the number of sets of configurations within a 5% tolerance compared to the best of these sets. Table 6 provides the population size, the number of generations and the number of times each operator is applied in the optimization of RSA using GAs. On the other hand, Table 7 provides the stopping criteria and other parameters used in the optimization of RSA using RTS as adopted from

Table 6 Parameters used in integer-coded GAs

Parameter	Value
Population size	140
Number of generations	120
Number of times of cross-over, application (arithmetic cross-over, simple cross-over, and heuristic cross-over)	Four times for each operator
Number of times of mutation application (uniform mutation, boundary mutation, non-uniform mutation, and whole non-uniform mutation)	Eight times for each operator

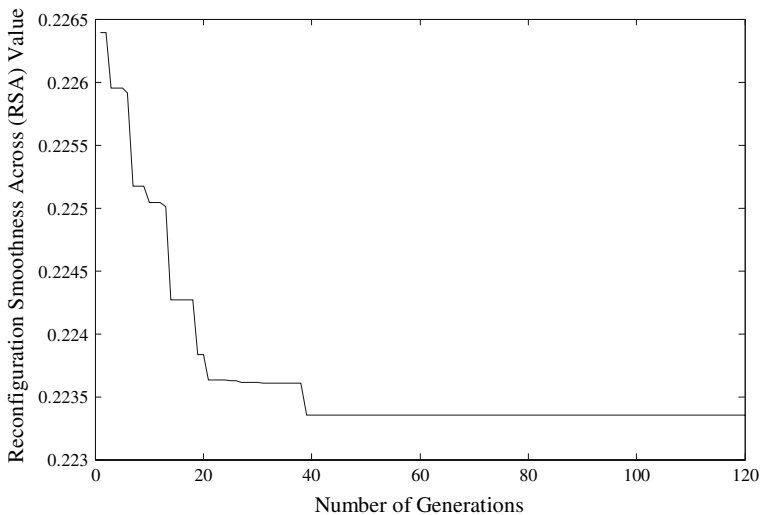
Table 7 Parameters used in RTS (Battiti and Techiolli 1994)

Parameter	Value
Maximum number of iterations	100
Maximum number of iterations with same best objective function value	50
Maximum number of solution encounters after which the solution is considered one of the often repeated solutions	Three times
Maximum number of often repeated solutions after which the escape mechanism is performed	Three times
Percentage by which the prohibition period is increased in diversification	1.1
Percentage by which the prohibition period is decreased in intensification	0.9

(Battiti and Techiolli 1994). RTS stops when either the number of iterations reaches 100 or 50 iterations passes without improving the best solution found so far.

5.2.1 Optimization results using integer-coded GAs

The integer-coded GAs generated 10 near-optimal sets of configuration, each of which is composed of a number of selected configurations corresponding to the different possible demand scenarios (DSs) that were identified in Table 3 corresponding to all the configuration periods (CPs) within the planning horizon of the RMS. The first two sets have the same near-optimal value of 0.22335 for the RSA. All the remaining sets were within the specified tolerance of 5% from the near-optimal value and have the same value of 0.22336 for the RSA, which is very close to the best achieved value. Figure 14 demonstrates the GA convergence curve for this run. The computation time required by the developed MATLAB toolbox in

**Fig. 14** The GA convergence curve for the second phase

this run to produce these solutions based on the presented optimization model was about 7.1 h.

Table 8 represents the first near-optimal set obtained in terms of the configurations selected from among those produced in the first phase. Figure 15 illustrates these configurations and Table 9 provides their performance characteristics, where C_{ij} is the configuration selected for DS_{ij} within the near-optimal set demonstrated in Table 8. The stage locations and reconfiguration smoothness (RS) evaluations for the configurations corresponding to CP1 and CP2 were determined using the reconfiguration planning procedure because there is only one possibility for their preceding configurations (C_0 prior to CP1 and C_{11} prior to CP2), whereas the stage locations for all the other anticipated future configurations could not be determined at this point in time because there exist more than one possibility for their preceding configurations.

Table 10 represents the second near-optimal set of selected configurations obtained, which has the same value of 0.22335 for the RSA as the first set. This set is identical to the first set except for C_{51} (the configuration corresponding to DS_{51}), which is configuration #1 in this set (the best configuration obtained for DS_{51}) rather than configuration #5 in the first set. This means that it is the same as C_{51} illustrated in Fig. 13 and reported in Table 5.

It is noticed that both sets have the same selected configuration for the first configuration period (CP1) which is configuration #3 from those generated by the first phase for CP1. Therefore C_1 (the configuration selected for CP1, the period of interest) is configuration #3 (C_{11} in Fig. 15 and Table 9) from those generated by the first phase for CP1.

The developed RMS Configuration Selection Approach provides the near-optimal sets of configurations, as well as the execution plans for reconfiguring the system from C_0 (Fig. 12) to C_1 of that set, which is configuration #3 for CP1 in this case (C_{11} in Fig. 15), according to the reconfiguration planning procedure.

Table 8 The first near-optimal set of selected configurations obtained by GAs

Configuration period	Demand scenario	Selected configuration
CP1	DS_{11}	Configuration #3
CP2	DS_{21}	Configuration #8
	DS_{22}	Configuration #6
CP3	DS_{31}	Configuration #3
	DS_{32}	Configuration #7
	DS_{33}	Configuration #8
CP4	DS_{41}	Configuration #4
	DS_{42}	Configuration #2
CP5	DS_{51}	Configuration #5
	DS_{52}	Configuration #7
	DS_{53}	Configuration #7
	DS_{54}	Configuration #8

C₁₁ (Configuration #3)							
S	1	2	3	4	5	6	7
(SL)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
M	1	1	1	1	1	1	2
MC	2	3	2	5	2	5	2
NMS	3	3	3	7	2	4	6
OS _A	1	0	14	5	0	3	6'
OS _B	1	15	13	5	11	9	6

C₂₁ (Configuration #8)							
S	1	2	3	4			
(SL)	(2)	(3)	(4)	(7)			
M	1	1	1	2			
MC	3	3	5	2			
NMS	1	2	6	5			
OS _A	1	15	5	6'			

C₂₂ (Configuration #6)								
S	1	2	3	4	5	6	7	
(SL)	(1)	(2)	(3)	(4)	(7)	(8)	(9)	
M	1	1	1	1	2	1	1	
MC	3	3	5	5	3	3	3	
NMS	2	3	4	7	5	2	2	
OS _A	1	15	0	5	6'	0	0	
OS _B	1	14	9	5	6	12	13	

C₃₁ (Configuration #3)								
S	1	2	3	4	5	6	7	8
M	1	1	1	2	1	2	1	1
MC	3	3	5	2	5	3	2	3
NMS	2	2	7	3	4	5	2	2
OS _A	1	2	5	0	7	6'	4	3
OS _B	1	14	5	3	9	6	11	13

C₃₂ (Configuration #7)							
S	1	2	3	4	5	6	7
M	1	1	1	1	1	1	2
MC	3	4	5	2	5	3	3
NMS	2	2	7	4	8	2	6
OS _A	0	0	1	15	5	0	6'
OS _B	1	14	5	12	9	13	6

C₃₃ (Configuration #8)							
S	1	2	3	4	5	6	
M	1	1	1	1	2	1	
MC	2	5	5	5	3	1	
NMS	2	7	3	5	5	2	
OS _B	1	17	9	5	6	11	

C₄₁ (Configuration #4)						
S	1	2	3	4	5	6
M	1	1	1	1	2	1
MC	2	5	5	5	3	3
NMS	2	5	4	5	5	2
OS _B	1	16	9	5	6	12

C₄₂ (Configuration #2)						
S	1	2	3	4	5	6
M	1	1	1	2	1	1
MC	2	5	5	3	5	3
NMS	2	6	6	4	3	2
OS _A	1	15	5	6'	0	0
OS _B	1	16	5	6	9	12

C₅₁ (Configuration #5)							
S	1	2	3	4	5	6	7
M	1	1	1	2	1	2	1
MC	3	5	5	4	5	3	2
NMS	2	7	4	2	7	4	2
OS _A	1	15	0	0	5	6'	0
OS _B	1	16	9	3	5	6	11

C₅₂ (Configuration #7)							
S	1	2	3	4	5	6	
M	1	1	1	1	1	2	
MC	2	5	5	2	5	3	
NMS	3	6	6	3	4	4	
OS _A	1	15	5	0	0	6'	
OS _B	1	16	5	12	9	6	

C₅₃ (Configuration #7)							
S	1	2	3	4	5	6	7
M	1	1	2	1	1	1	2
MC	2	5	3	5	5	2	2
NMS	2	6	3	7	5	2	6
OS _A	0	1	0	15	5	0	6'
OS _B	1	16	3	5	9	11	6

C₅₄ (Configuration #8)				
S	1	2	3	4
M	1	1	1	2
MC	2	2	5	3
NMS	2	3	6	3
OS _A	1	15	5	6'

Fig. 15 The first near-optimal set of selected configurations obtained by GAs

The step-by-step action plan for reconfiguring the RMS from C0 to C1 is as follows:

(I) Stages:

1. Stage of type M1 located in SL3 keeps its location in the system.
2. Stage of type M1 located in SL4 keeps its location in the system.
3. Stage of type M1 located in SL5 keeps its location in the system.
4. Stage of type M2 located in SL7 keeps its location in the system.

Table 9 Characteristics for the first near-optimal set of selected configurations obtained by GAs

C (C #)	Capital cost (in millions of US dollars)	Initial investment (in millions of US dollars)	RS	System availability (in %)	Expected production rate for Part A (in parts/h)	Expected production rate for Part B (in parts/h)	System utilization (in %)
C ₁₁ (#3)	7.7802	27.82	0.29392	73.526	343	284	98.3
C ₂₁ (#8)	2.5722	13.095	0.12808	66.764	247	N/A	89.2
C ₂₂ (#6)	5.4047	27.515	0.1785	63.11	350	269	95.9
C ₃₁ (#3)	6.2782	27.2	N/A	79.815	342	280	99.3
C ₃₂ (#7)	7.6493	33.14	N/A	75.934	399	333	99.2
C ₃₃ (#8)	5.2569	22.775	N/A	54.2	N/A	203	98.4
C ₄₁ (#4)	6.4	22.885	N/A	60.776	N/A	221	99.6
C ₄₂ (#2)	6.4798	23.17	N/A	59.831	284	214	98.4
C ₅₁ (#5)	6.9263	27.99	N/A	71.816	336	275	99.1
C ₅₂ (#7)	6.4091	25.9	N/A	66.571	296	255	97.7
C ₅₃ (#7)	6.9894	28.245	N/A	58.594	269	279	98.4
C ₅₄ (#8)	3.4483	13.935	N/A	38.128	266	N/A	93.8

Table 10 The second near-optimal set of selected configurations obtained by GAs

Configuration period	Demand scenario	Selected configuration
CP1	DS ₁₁	Configuration #3
CP2	DS ₂₁	Configuration #8
	DS ₂₂	Configuration #6
CP3	DS ₃₁	Configuration #3
	DS ₃₂	Configuration #7
	DS ₃₃	Configuration #8
CP4	DS ₄₁	Configuration #4
	DS ₄₂	Configuration #2
CP5	DS ₅₁	Configuration #1
	DS ₅₂	Configuration #7
	DS ₅₃	Configuration #7
	DS ₅₄	Configuration #8

5. Stage of type M2 located in SL6 is totally removed from the system.
 6. Stage of type M1 located in SL8 is relocated to SL1 of the system.
 7. A new stage of type M1 is added to SL2 of the system.
 8. A new stage of type M1 is added to SL6 of the system.
- (II) Machines:
1. 3 M1 machines located in SL3 keep their location in the system.
 2. 6 M1 machines located in SL4 keep their location in the system.

3. 2 M1 machines located in SL5 keep their location in the system.
3. 2 M1 machines located in SL5 keep their location in the system.
4. 6 M2 machines located in SL7 keep their location in the system.
5. 2 M2 machines located in SL6 are totally removed from the system.
6. 1 M1 machine located in SL3 is relocated to SL2 of the system.
7. 2 M1 machines located in SL5 are relocated to SL2 of the system.
8. 1 M1 machine located in SL5 is relocated to SL6 of the system.
9. 2 M1 machines located in SL8 are relocated to SL1 of the system.
10. 1 new M1 machine is added to SL1 of the system.
11. 1 new M1 machine is added to SL4 of the system.
12. 3 new M1 machines are added to SL6 of the system.

5.2.2 Optimization results using RTS

The Reactive Tabu Search (RTS) was applied to the same outcomes of the first phase that were used for the GAs optimization in order to be able to compare the results. RTS generated 10 near-optimal sets of configurations, each of which is composed of a number of selected configurations corresponding to the different possible demand scenarios (DSs) that were identified in Table 3 corresponding to all the configuration periods (CPs) within the planning horizon of the RMS. The first two sets have the same near-optimal value of 0.22335 for the RSA like GAs and are the same exact solutions (sets of configurations) (Tables 8, 10). All the remaining sets were within the specified tolerance of 5% from the near-optimal value and have the same value of 0.22336 for the RSA, which is very close to the best-achieved value. Figure 16 demonstrates the RTS convergence curve for this run. The

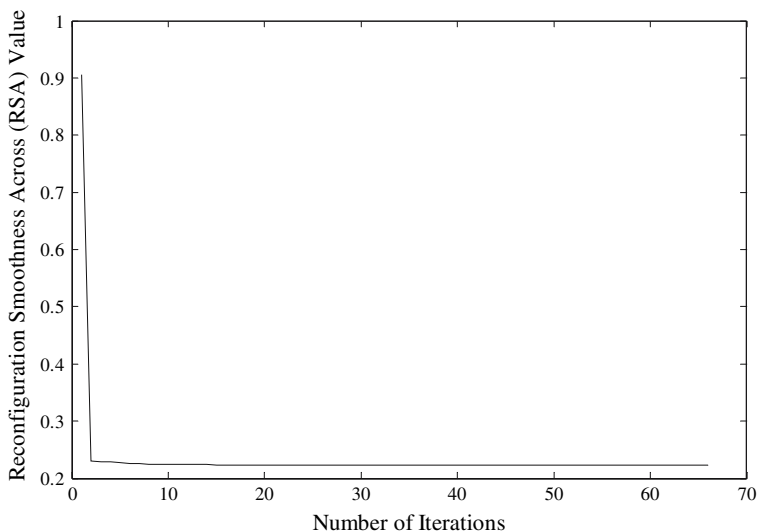


Fig. 16 The RTS convergence curve for the second phase

computation time required by the developed MATLAB toolbox in this run to produce these solutions based on the presented optimization model was about 4.8 h. The usual time/run for the RTS with these parameters would on average take more time but this run stopped before reaching 100 iterations because the other stopping criterion was satisfied (no improvement in the best solution found so far for 50 iterations) which lead to stopping after just 67 iterations.

Both sets, which are identical to those obtained by GAs, have the same selected configuration for the first configuration period (CP1) which is configuration #3 from those generated by the first phase for CP1. Therefore C1 (the configuration selected for CP1, the period of interest) is configuration #3 (C_{11} in Fig. 15 and Table 9) from those generated by the first phase for CP1. Accordingly the same reconfiguration execution plan from C0 to C1 applies.

5.3 Discussion of results

The results obtained in the first phase, as sampled in Figs. 13 and 15 and Tables 5 and 9, show that the first phase of the proposed approach succeeded in generating configurations that are not only of good quality, in terms of cost and availability, but also provide almost exactly the capacity needed when needed as the system utilization for all the generated configurations is very close to 100%. In addition, the equipment (machines and their configurations) selected within each configuration and the number of production stages reflect the functionality requirements (parts being produced) with no redundancy in stage allocations, which means that the generated configurations provide the functionality needed when it is needed.

The results showed that the selected configuration for CP1 in the case study was the third ordered configuration from those produced in the first phase in terms of its cost and availability. Also the tentative selection of configurations for future periods confirms that same concept. This proves that the system configuration with best performance in a specific period may not be the best configuration for that period if the reconfiguration effort is considered over the planning horizon of the system. When comparing the performance characteristics of that selected configuration for CP1 (C_{11} in Fig. 15 and Table 9) with the best obtained configuration for the same period CP1 (C_{11} in Fig. 13 and Table 5), it is clear that the increase in capital cost is less than 1% (5.6 thousand US Dollars) and the decrease in the system availability is also less than 1% (0.635%). This means that the developed approach managed to select a configuration that minimizes the effort of reconfiguration along the planning horizon without sacrificing the first phase performance objectives of minimizing the cost and maximizing the system availability. This is because the optimization model adopted in the first phase of the RMS Configuration Selection Approach is capable of producing more than one near-optimal alternate configuration, which provides flexibility in the selection process if other criteria are being considered, such as the RSA, in the second phase. In addition, the predefined maximum tolerance (e.g. 5% or any other value), which limits the number of configurations that qualify to the second phase of the approach, warrants that the

near-optimal configuration sets selected in the second phase are within an acceptable range (5% or other), in terms of their evaluation criteria, from the best obtained configurations. This guarantees that the minimization of the RSA value is performed without sacrificing the first phase performance objectives.

It is noticeable from comparing the outcomes of optimization and the performance of both optimization techniques, GAs and RTS, in the second phase that the final results were consistent, which increases confidence in the optimization model developed for the second phase and its results.

Generally, and from the outcomes of many other various runs, both techniques used in the second phase are usually consistent in reaching the same near-optimal values for RSA, which indicates that they might be reaching global optimal values. Both techniques have, on average, the same time/run except if the second stopping criterion of RTS was triggered, which gives better time/run for RTS. In most of the runs, RTS proved to be more powerful in terms of its capability to produce multiple near-optimal solutions with same objective function value, the best value arrived at, which is very useful for the decision maker to have different alternatives to choose from.

6 Conclusions and future work

Selection of manufacturing systems configurations is essential for RMSs to adapt to dynamic market changes throughout their life cycle where the reconfiguration capabilities widen the possibilities of system level configurations. A thorough literature review of the current state of research regarding RMS configuration selection has been provided. However, there remain many challenges in this area of research. A novel RMS Configuration Selection Approach that addresses most of the shortcomings of the previous work was proposed. It considers more than one aspect of the system configuration (arrangement of machines, equipment selection and assignment of operations), utilizes important system-level evaluation criteria (cost and availability) and involves stochastic analysis to take into consideration the smoothness of the anticipated reconfiguration process. A new discrete optimization procedure was introduced in the second phase for selecting sets of optimal configurations, from those produced in the first phase, to optimize the RSA level over the planning horizon. Integer-coded GAs and RTS capable of producing multiple near-optimal solutions have been successfully implemented.

The proposed algorithms were implemented using a toolbox, RMS-Configurator, developed using MATLAB software. A case study was presented and the results obtained using both GAs and RTS were consistent, which validates the developed approach. It was demonstrated that it is possible within a RMS environment to provide the capacity and functionality needed when needed with the least amount of reconfiguration effort without sacrificing the targeted system performance objectives. The results showed that the smoothness of subsequent reconfiguration processes is influenced by the initial manufacturing systems configuration selection,

and that the system configuration with best performance in a specific period may not be the best configuration for that period if the reconfiguration effort is considered for the entire planning horizon of the system.

In practice, the proposed RMS Configuration Selection Approach would be applied at the initial design stage of the manufacturing system or when reconfiguration is considered at the beginning of each configuration period. The best quality of demand forecast and future data should be used in the analysis to determine the configuration for the next period, from those generated in the first phase, with the best reconfiguration smoothness. This process is repeated using updated expected demand information whenever reconfiguration is contemplated. This iterative application of the proposed approach ensures that the configuration transition cost and effort over the expected life time of the manufacturing system is minimized.

This research work enhances the existing knowledge about manufacturing systems with regards to configurations selection by taking into consideration the anticipated level of reconfiguration effort over the anticipated planning horizon. This helps support the management decisions regarding system configurations for a reconfigurable manufacturing system by providing the best alternatives at the beginning of each configuration period.

The proposed approach may be extended by considering additional aspects of the RMS configuration structures, such as the material handling systems to be used with different types of equipment. In addition, system evaluation criteria other than cost and system availability may be considered within a multi-objective optimization framework.

Acknowledgements The contributions from the Canada Research Chairs (CRC) program and the Natural Sciences and Engineering Research Council (NSERC) of Canada in support of this research are greatly acknowledged.

Appendix A. Optimization using Reactive Tabu Search (RTS)

Basic Tabu Search (TS)

TS is a kind of iterative search characterized by the use of a flexible memory. It is able to eliminate local minima and to find the global minimum of a multi-modal search space. The process with which TS overcomes the local optimality problem is based on moving to the best (most improved or least deteriorated) admissible solution in the neighborhood of the current solution in terms of the objective value and tabu restrictions at each iteration. A tabu list is employed to store the characteristics of accepted moves so that these characteristics can be used to classify certain moves as tabu (i.e. to be avoided) in later iterations. Since moves not leading to improvements are accepted in TS, it is possible to return to already-visited solutions. This might cause a cycling problem to arise. The tabu list is used to overcome this problem. A strategy called the forbidding strategy is employed to

control and update the tabu list. By using the forbidding strategy, a path previously visited is avoided and new regions of search space are explored. On the other hand, an aspiration criterion is used to make a tabu solution free if this solution is of sufficient quality and can prevent cycling. Two highly important components of TS are intensification and diversification strategies. The main difference between intensification and diversification is that during intensification stage the search focuses on examining neighbors of elite solutions. The diversification, on the other hand, encourages the search process to examine unvisited regions and generate new solutions.

Reactive Tabu Search (RTS)

The Reactive Tabu Search (RTS) (Battiti and Tecchiolli 1994) is based on the idea of dynamic implementation of the intensification and diversification strategies throughout the search process using mechanisms that are function of the search outcomes rather than fixed mechanisms. These mechanisms dynamically resize the tabu list size according to the search dynamics, which is being followed by making use of memory. The tabu list size in RTS increases to achieve diversification in areas of the search where solutions are re-visited more frequently. On the other hand, intensification is performed by decreasing the tabu list size in areas of promising objective function evaluations and where the solutions are re-visited less frequently. In addition, an escape mechanism that performs a number of totally random moves is devised in order to escape from being trapped in large basins of the solution space where the number of frequently visited solutions exceeds a specific limit. RTS deals efficiently with combinatorial optimization problems.

Appendix B. Example parts machine processing information

Table B1 Operations data for part ANC-90

Feature	Description	Operation	Op. ID	TAD candidates	Tool candidates
F1	Planar surface	Milling	OP1	+Z	C6, C7, C8
F2	Planar surface	Milling	OP2	−Z	C6, C7, C8
F3	Four holes arranged as a replicated feature	Drilling	OP3	+Z, −Z	C2
F4	A step	Milling	OP4	+X, −Z	C6, C7
F5	A protrusion (rib)	Milling	OP5	+Y, −Z	C7, C8
F6	A protrusion	Milling	OP6	−Y, −Z	C7, C8

Table B1 continued

Feature	Description	Operation	Op. ID	TAD candidates	Tool candidates
F7	A compound hole	Drilling	OP7	-Z	C2, C3, C4
		Reaming	OP8		C9
		Boring	OP9		C10
F8'	Six holes arranged in a replicated feature	Drilling	OP10'	-Z	C1
		Tapping	OP11'		C5
F9	A step	Milling	OP12	-X, -Z	C6, C7

Table B2 Operations data for part ANC-101

Feature	Description	Operation	Op. ID	TAD candidates	Tool candidates
F1	Planar surface	Milling	OP1	+Z	C6, C7, C8
F2	Planar surface	Milling	OP2	-Z	C6, C7, C8
F3	Four holes arranged as a replicated feature	Drilling	OP3	+Z, -Z	C2
F4	A step	Milling	OP4	+X, -Z	C6, C7
F5	A protrusion (rib)	Milling	OP5	+Y, -Z	C7, C8
F6	A protrusion	Milling	OP6	-Y, -Z	C7, C8
F7	A compound hole	Drilling	OP7	-Z	C2, C3, C4
		Reaming	OP8		C9
		Boring	OP9		C10
F8	Nine holes arranged in a replicated feature	Drilling	OP10	-Z	C1
		Tapping	OP11		C5
F9	A step	Milling	OP12	-X, -Z	C6, C7
F10	Two pockets arranged as a replicated feature	Milling	OP13	+X	C6, C7, C8
F11	A boss	Milling	OP14	-a	C7, C8
F12	A compound hole	Drilling	OP15	-a	C2, C3, C4
		Reaming	OP16		C9
		Boring	OP17		C10
F13	A pocket	Milling	OP18	-X	C7, C8
F14	A compound hole	Reaming	OP19	+Z	C9
		Boring	OP20		C10

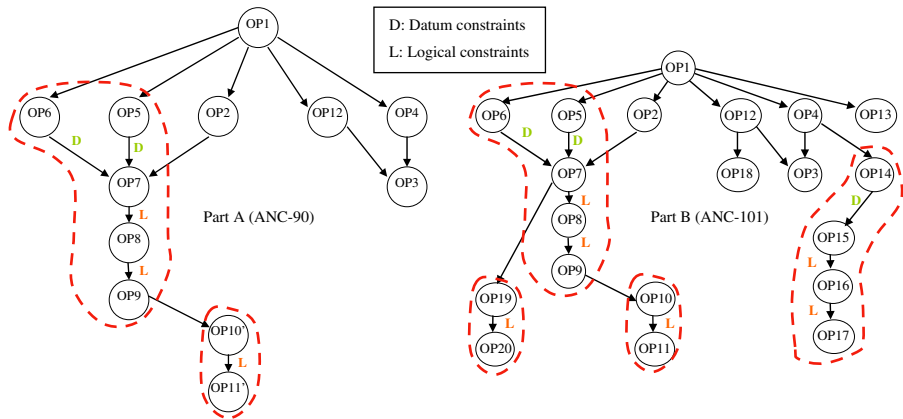


Fig. B1 Operations precedence graph for the two parts

Table B3 Operation clusters definitions for part ANC-90

Operation cluster	Operations
OC1	[OP1]
OC2	[OP2]
OC3	[OP3]
OC4	[OP4]
OC5	[OP5, OP6, OP7, OP8, OP9]
OC6'	[OP10', OP11']
OC7	[OP12]

Table B4 Operation clusters definitions for part ANC-101

Operation cluster	Operations
OC1	[OP1]
OC2	[OP2]
OC3	[OP3]
OC4	[OP4]
OC5	[OP5, OP6, OP7, OP8, OP9]
OC6	[OP10, OP11]
OC7	[OP12]
OC8	[OP13]
OC9	[OP14, OP15, OP16, OP17]
OC10	[OP18]
OC11	[OP19, OP20]

Table B5 Time and production rate information for different M-MC-OS combinations

Operation clusters setup (OS)		Standard time in seconds (production rate in parts/h)									
Code	Operation clusters (OCs)	M1					M2				
		MC1 ₁	MC1 ₂	MC1 ₃	MC1 ₄	MC1 ₅	MC2 ₁	MC2 ₂	MC2 ₃	MC2 ₄	
OS1	[OC1]	30 (120)	30 (240)	30 (360)	30 (480)	30 (120)	X	X	X	X	
OS2	[OC2]	20 (180)	20 (360)	20 (540)	20 (720)	20 (180)	X	X	X	X	
OS3	[OC3]	30 (120)	30 (240)	30 (360)	30 (480)	30 (120)	30 (120)	30 (240)	30 (360)	30 (480)	
OS4	[OC4]	20 (180)	20 (360)	20 (540)	20 (720)	20 (180)	X	X	X	X	
OS5	[OC5]	X	X	X	X	60 (60)	X	X	X	X	
OS6	[OC6]	120 (30)	120 (60)	120 (90)	120 (120)	120 (30)	120 (30)	120 (60)	120 (90)	120 (120)	
OS6'	[OC6']	90 (40)	90 (80)	90 (120)	90 (160)	90 (40)	90 (40)	90 (80)	90 (120)	90 (160)	
OS7	[OC7]	18 (200)	18 (400)	18 (600)	18 (800)	18 (200)	X	X	X	X	
OS8	[OC8]	X	X	X	X	20 (180)	X	X	X	X	
OS9	[OC9]	X	X	X	X	40 (90)	X	X	X	X	
OS10	[OC10]	X	X	X	X	18 (200)	X	X	X	X	
OS11	[OC11]	24 (150)	24 (300)	24 (450)	24 (600)	24 (150)	X	X	X	X	
OS12	[OC3, OC11]	60 (60)	60 (120)	60 (180)	60 (240)	60 (60)	X	X	X	X	
OS13	[OC8, OC10]	30 (120)	30 (240)	30 (360)	30 (480)	30 (120)	X	X	X	X	
OS14	[OC2, OC4, OC7]	40 (90)	40 (180)	40 (270)	40 (360)	40 (90)	X	X	X	X	
OS15	[OC2, OC3, OC4, OC7]	60 (60)	60 (120)	60 (180)	60 (240)	60 (60)	X	X	X	X	
OS16	[OC2, OC4, OC7, OC8, OC10]	X	X	X	X	60 (60)	X	X	X	X	
OS17	[OC2, OC3, OC4, OC7, OC8, OC10]	X	X	X	X	90 (40)	X	X	X	X	

References

- Abdi MR, Labib AW (2003) A design strategy for reconfigurable manufacturing systems (RMSs) using analytical hierarchical process (AHP): a case study. *Int J Prod Res* 41(10):2273–2299
- Abdi MR, Labib AW (2004) Grouping and selecting products: the design key of Reconfigurable Manufacturing Systems (RMSs). *Int J Prod Res* 43(3):521–546
- Battiti R, Tecchiolli G (1994) The Reactive Tabu Search. *ORSA J Comput* 6:126–140
- Cochran DS, Arinez JF, Duda JW, Linck J (2001) A Decomposition Approach for Manufacturing System Design. *J Manuf Syst* 20(6):371–389
- ElMaraghy HA (2005) Flexible and Reconfigurable Manufacturing Systems paradigms. *Int J Flex Manuf Syst* 17(4):261–276 (Special issue on Reconfigurable Manufacturing Systems)
- Fukaya T (2004) HIPARMS (Highly Productive and Reconfigurable Manufacturing System) Final Report, <http://www.ims.org/images/projects/hiparms/finalreport.pdf>
- Glover F (1986) Future paths for integer programming and links to artificial intelligence. *Comput Operat Res* 13:533–549
- Glover F, Laguna M (1997) *Tabu Search*. Kluwer Academic Publishers, Boston
- Herrera F, Lozano M, Verdegay JL (1998) Tackling real-coded Genetic Algorithms: operators and tools for behavioral analysis. *Artif Intell Rev* 12(4):265–319
- Holland J (1975) *Adaptation in natural and artificial systems*. University of Michigan Press, Ann Arbor, MI
- IEEE Std 1320.1 (1998) IEEE Standard for Functional Modeling Language—Syntax and Semantics for IDEF0, IEEE, New York
- Kimms A (2000) Minimal investment budgets for flow line configuration. *IIE Trans* 32:287–298
- Koren Y, Hu SJ, Weber TW (1998) Impact of manufacturing system configuration on performance. *Ann CIRP* 47(1):369–372
- Koren Y, Heisel U, Jovane F, Moriwaki T, Pritschow G, Ulsoy G, Van Brussel H (1999) Reconfigurable manufacturing systems. *Ann CIRP* 48(2):527–540
- Kuo CH (2001) Resource allocation and performance evaluation of the Reconfigurable Manufacturing Systems. *Proc IEEE Int Conf Syst Man Cybernet* 4:2451–2456
- Lee GH (1997) Reconfigurability consideration design of components and manufacturing systems. *Int J Adv Manuf Technol* 13(5):376–386
- Lee GH (1998) Designs of components and manufacturing systems for agile manufacturing. *Int J Prod Res* 36(4):1023–1044
- Maier-Sperdelozzi V, Hu SJ (2002) Selecting manufacturing system configurations based on performance using AHP. Technical Paper—Society of Manufacturing Engineers. MS, no MS02-179, pp 1–8
- Maier-Sperdelozzi V, Koren Y, Hu SJ (2003) Convertibility measures for manufacturing systems. *Ann CIRP* 52(1):367–370
- Makino H, Trai T (1994) New developments in assembly systems. *Ann CIRP* 43(2):501–522
- Matta A, Tolio T, Karaesmen F, Dallery Y (2001) An integrated approach for the configuration of automated manufacturing systems. *Robot Comput Integ Manuf* 17:19–26
- Mehrabi MG, Ulsoy AG, Koren Y (2000) Reconfigurable manufacturing systems: key to future manufacturing. *J Intell Manuf* 11:403–419
- Ohiro T, Myreshka, Morikawa K, Takahashi K (2003) A stochastic model for deciding an optimal production order and its corresponding configuration in a reconfigurable manufacturing system with multiple product groups. *Proceedings of the CIRP 2nd international conference on reconfigurable manufacturing*, Michigan
- Son SY (2000) *Design principles and methodologies for reconfigurable machining systems*. PhD Dissertation, University of Michigan
- Spicer JP (2002) *A design methodology for scalable machining systems*. PhD Dissertation, University of Michigan
- Spicer P, Koren Y, Shpitalni M, Yip-Hoi D (2002) Design principles for machining system configurations. *Ann CIRP* 51(1):275–280
- Tang L, Yip-Hoi DM, Wang W, Koren Y (2004) Concurrent line-balancing, equipment selection and throughput analysis for multi-part optimal line design. *Int J Manuf Sci Prod* 6(1–2):71–81

- Tang L, Yip-Hoi DM, Wang W, Koren Y (2005) Selection principles on manufacturing system for part family. Proceedings of the CIRP 3rd international conference on reconfigurable manufacturing, Ann Arbor, MI
- Tesfamariam D (2005) Configuration design of a high performance and responsive manufacturing system. PhD Dissertation, KTH Royal Institute of Technology, Stockholm, Sweden
- Toenshoff HK, Schnuelle A, Rietz W (2001) Planning of manufacturing systems with methods analogous to nature. International CIRP design seminar, Stockholm, pp 255–260
- Ushakov IA (1986) A universal generating function. *Soviet J Comput Syst Sci* 24(5):118–129
- Wiendahl H-P, Heger CL (2003) Justifying changeability: a methodical approach to achieving cost effectiveness. Proceedings of the CIRP 2nd international conference on reconfigurable manufacturing, Michigan
- Xiaobo Z, Jiancai W, Zhenbi L (2000a) A stochastic model of a reconfigurable manufacturing system. Part 1. A framework. *Int J Prod Res* 38(10):2273–2285
- Xiaobo Z, Wang J, Luo Z (2000b) A stochastic model of a reconfigurable manufacturing system. Part 2. Optimal configurations. *Int J Prod Res* 38(12):2829–2842
- Xiaobo Z, Wang J, Luo Z (2001a) A stochastic model of a reconfigurable manufacturing system. Part 3. Optimal selection policy. *Int J Prod Res* 39(4):747–758
- Xiaobo Z, Wang J, Luo Z (2001b) A stochastic model of a reconfigurable manufacturing system. Part 4. Performance measure. *Int J Prod Res* 39(6):1113–1126
- Yamada Y, Ookoudo K, Komura Y (2003) Layout optimization of manufacturing cells and allocation optimization of transport robots in Reconfigurable Manufacturing Systems using particle swarm optimization. *IEEE Int Conf Intell Robots Syst* 2:2049–2054
- Yang S, Hu SJ (2000) Productivity analysis of a six CNC machine manufacturing system with different configurations. Proceedings of the 2000 Japan–USA flexible automation conference, Michigan, July, pp 499–505
- Youssef AMA (2006) Optimal configuration selection for reconfigurable manufacturing systems. PhD Dissertation, University of Windsor, Canada
- Youssef AMA, ElMaraghy HA (2006a) Assessment of manufacturing systems reconfiguration smoothness. *Int J Adv Manuf Technol* 30(1–2):174–193
- Youssef AMA, ElMaraghy HA (2006b) Modelling and optimization of multiple-aspect rms configurations. *Int J Prod Res* 44(22):4929–4958 (Special issue on advances in evolutionary computation for design and manufacturing problems)
- Youssef AMA, Mohib A, ElMaraghy HA (2006) Availability assessment of multi-state manufacturing systems using universal generating function. *Ann CIRP* 55(1):445–448
- Youssef AMA, ElMaraghy HA (2007) Availability consideration in the optimal selection of multiple-aspect RMS configurations. *Int J Prod Res* (in press) Manuscript ID is TPRS-226074
- Zhong W, Maier-Sperdelozzi V, Bratzel A, Yang S, Chick SE, Hu SJ (2000) Performance analysis of machining systems with different configurations. Proceedings of the 2000 Japan–USA flexible automation conference, Michigan, July, pp 783–790



Ayman M.A. Youssef, Ph.D. is a Post Doctoral Fellow in the Intelligent Manufacturing Systems (IMS) Centre at the University of Windsor, Canada. He is currently on a leave of absence from his original position as an Assistant Lecturer in the Mechanical Design and Production Engineering Department, Faculty of Engineering at Cairo University, Egypt. His research interests are in manufacturing systems optimal design, modelling, performance analysis and reconfiguration. Dr. Ayman Youssef received his B.Sc. and M.Sc. degrees in Mechanical Engineering from Cairo University (1998, 2001) and his Ph.D. degree in Manufacturing Systems Engineering from the University of Windsor (2006).



Hoda A. ElMaraghy, Ph.D., P.Eng. is currently Canada Research Chair in Manufacturing Systems, Professor of Industrial and Manufacturing Systems Engineering and Director of the Intelligent Manufacturing Systems (IMS) Centre at the University of Windsor. She is a Fellow of the Canadian Society of Mechanical Engineers (CSME), Fellow of the Society of Manufacturing Engineers (SME), and Fellow of the International Academy for Production Engineering (CIRP) and Member of the American Society of Mechanical Engineers (ASME). Her teaching and research interests are in manufacturing systems design, modelling, flexibility, reconfiguration, control, quality, inspection, assembly, process planning and robotics. Dr. Hoda ElMaraghy published more than 350 papers in 6 scientific and professional journals and conferences, edited 2 books

and contributed to 13 other books. She is a member of Editorial Boards of four International Scientific and Engineering Journals. She received a mechanical engineering Bachelor degree from Cairo University, Egypt and Master's and Ph.D. degrees from McMaster University, Canada where she became a Professor and founding Director of the Flexible Manufacturing Systems Centre until joining the University of Windsor as the Dean of Engineering in 1994.